

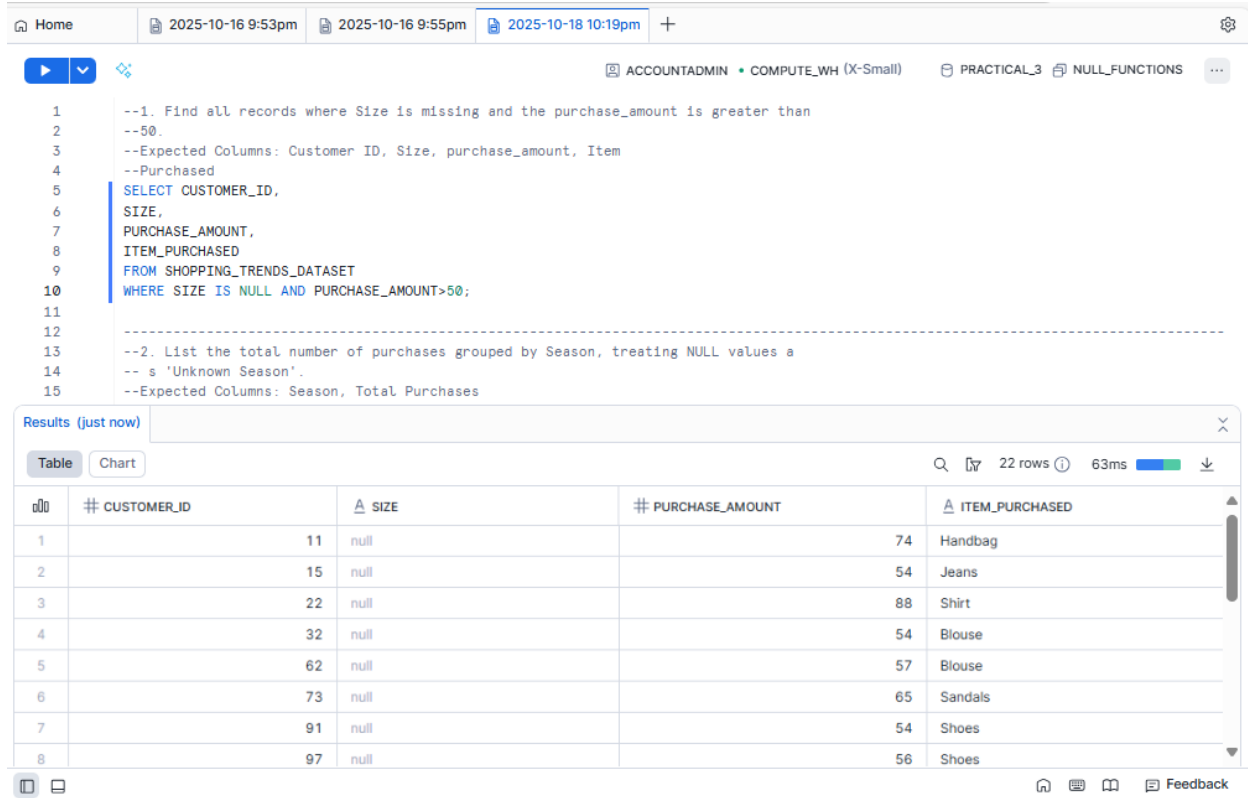
BrightLight Data Analytics Coding Practical

ADVANCE SQL

PRACTICAL-3

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1. Find all records where Size is missing and the purchase_amount is greater than 50. Expected Columns: Customer ID, Size, purchase_amount, Item Purchased



The screenshot shows a SQL IDE interface with a query editor and a results pane. The query editor contains two SQL queries. The first query is a SELECT statement that filters records where the size is null and the purchase amount is greater than 50. The results pane shows 22 rows of data, with the first 8 rows displayed. The columns are CUSTOMER_ID, SIZE, PURCHASE_AMOUNT, and ITEM_PURCHASED.

```
1 --1. Find all records where Size is missing and the purchase_amount is greater than
2 --50.
3 --Expected Columns: Customer ID, Size, purchase_amount, Item
4 --Purchased
5 SELECT CUSTOMER_ID,
6 SIZE,
7 PURCHASE_AMOUNT,
8 ITEM_PURCHASED
9 FROM SHOPPING_TRENDS_DATASET
10 WHERE SIZE IS NULL AND PURCHASE_AMOUNT>50;
11
12 -----
13 --2. List the total number of purchases grouped by Season, treating NULL values a
14 -- s 'Unknown Season'.
15 --Expected Columns: Season, Total Purchases
```

Results (just now)

#	CUSTOMER_ID	SIZE	PURCHASE_AMOUNT	ITEM_PURCHASED
1	11	null	74	Handbag
2	15	null	54	Jeans
3	22	null	88	Shirt
4	32	null	54	Blouse
5	62	null	57	Blouse
6	73	null	65	Sandals
7	91	null	54	Shoes
8	97	null	56	Shoes

2. List the total number of purchases grouped by Season, treating NULL values a s 'Unknown Season'. Expected Columns: Season, Total Purchases

Home | 2025-10-16 9:53pm | 2025-10-16 9:55pm | 2025-10-18 10:19pm | +

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```

12
13 --2. List the total number of purchases grouped by Season, treating NULL values as
14 -- s 'Unknown Season'.
15 --Expected Columns: Season, Total Purchases
16 SELECT COALESCE(SEASON, 'UNKNOWN SEASON') AS SEASON,
17 COUNT(*) AS TOTAL_PURCHASES
18 FROM SHOPPING_TRENDS_DATASET
19 GROUP BY SEASON;
20
21
22
23
24
25
26 --3. Count how many customers used each Payment Method, treating NULLs as

```

Results (3 minutes ago)

Table Chart 22 rows 63ms

#	CUSTOMER_ID	SIZE	PURCHASE_AMOUNT	ITEM_PURCHASED
1	11	null	74	Handbag
2	15	null	54	Jeans
3	22	null	88	Shirt
4	32	null	54	Blouse
5	62	null	57	Blouse
6	73	null	65	Sandals
7	91	null	54	Shoes
8	97	null	56	Shoes

Feedback

3. Count how many customers used each Payment Method, treating NULLs as 'Not Provided'. Expected Columns: Payment Method, Customer Count

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```

25
26 --3. Count how many customers used each Payment Method, treating NULLs as
27 --'Not Provided'.
28 --Expected Columns: Payment Method, Customer Count
29 SELECT COALESCE(PAYMENT_METHOD, 'NOT PROVIDED') AS PAYMENT_METHOD,
30 COUNT(CUSTOMER_ID) AS CUSTOMER_COUNT,
31 FROM SHOPPING_TRENDS_DATASET
32 GROUP BY PAYMENT_METHOD;
33
34
35
36
37
38
39 --4. Show customers where Promo Code Used is NULL and Review Rating is below

```

Results (just now)

Table Chart 7 rows 81ms

#	PAYMENT_METHOD	CUSTOMER_COUNT
1	Credit Card	44
2	PayPal	51
3	Debit Card	42
4	NOT PROVIDED	30
5	Cash	42
6	Bank Transfer	38
7	Venmo	53

Feedback

4. Show customers where Promo Code Used is NULL and Review Rating is below 3.0. Expected Columns: Customer ID, Promo Code Used, Review Rating, Item Purchased

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```

37
38
39 --4. Show customers where Promo Code Used is NULL and Review Rating is below
40 --3.0.
41 --Expected Columns: Customer ID, Promo Code Used, Review Rating, Item Purchased
42
43 SELECT CUSTOMER_ID,
44        PROMO_CODE_USED,
45        REVIEW_RATING,
46        ITEM_PURCHASED,
47 FROM SHOPPING_TRENDS_DATASET
48 WHERE PROMO_CODE_USED IS NULL
49 AND REVIEW_RATING<3.0;
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65

```

Results (just now)

Table Chart 8 rows 70ms

	# CUSTOMER_ID	0 1 PROMO_CODE_USED	# REVIEW_RATING	ITEM_PURCHASED
1	21	null	2.5	Jeans
2	38	null	2.6	Jeans
3	61	null	2.5	Jeans
4	80	null	2.6	Sneakers
5	125	null	2.8	Sneakers
6	128	null	2.5	Shoes
7	180	null	2.5	Shorts
8	285	null	2.9	Blouse

Feedback

5. Group customers by Shipping Type, and return the average purchase_amount, treating missing values as 0. Expected Columns: Shipping Type, Average purchase_amount

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```

50
51
52 --5. Group customers by Shipping
53 --Type, and return the average purchase_amount, treating missing values as 0.
54 --Expected Columns: Shipping Type, Average purchase_amount
55
56 SELECT SHIPPING_TYPE,
57        AVG(PURCHASE_AMOUNT) AS AVERAGE_PURCHASE_AMOUNT
58 FROM SHOPPING_TRENDS_DATASET
59 GROUP BY SHIPPING_TYPE;
60
61
62
63
64
65

```

Results (1 minute ago)

Table Chart 8 rows 70ms

	# CUSTOMER_ID	0 1 PROMO_CODE_USED	# REVIEW_RATING	ITEM_PURCHASED
1	21	null	2.5	Jeans
2	38	null	2.6	Jeans
3	61	null	2.5	Jeans
4	80	null	2.6	Sneakers
5	125	null	2.8	Sneakers
6	128	null	2.5	Shoes
7	180	null	2.5	Shorts
8	285	null	2.9	Blouse

Feedback

6. Display the number of purchases per Location only for those with more than 5 purchases and no NULL Payment Method. Expected Columns: Location, Total Purchases

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2025-10-16 9:53pm
2025-10-16 9:55pm
2025-10-18 10:19pm
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COMPUTE_WH (X-Small)
PRACTICAL_3
NULL_FUNCTIONS
...

```

63
64
65
66
67 --6. Display the number of purchases per Location only for those with more than
68 --5 purchases and no NULL Payment Method.
69 --Expected Columns: Location, Total Purchases
70 SELECT LOCATION,
71 COUNT (ITEM_PURCHASED) AS TOTAL_PURCHASES
72 FROM SHOPPING_TRENDS_DATASET
73 WHERE PAYMENT_METHOD IS NOT NULL
74 GROUP BY LOCATION
75 HAVING COUNT (*) > 5;
76
77

```

Results (just now)

Table
Chart
9 rows
86ms

	LOCATION	TOTAL_PURCHASES
1	Maine	36
2	Kentucky	28
3	null	20
4	New York	28
5	Oregon	26
6	Rhode Island	27
7	Florida	31
8	Massachusetts	31

Feedback

7. Create a column Spender Category that classifies customers using CASE: 'High' if amount > 80, 'Medium' if BETWEEN 50 AND 80, 'Low' otherwise. Replace NULLs in purchase_amount with 0. Expected Columns: Customer ID, purchase_amount, Spender Category

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```

80
81
82 --7. Create a column Spender Category that classifies customers using CASE:
83 --'High' if amount > 80, 'Medium' if BETWEEN 50 AND 80,
84 --'Low' otherwise. Replace NULLs in purchase_amount with 0.
85 --Expected Columns: Customer ID, purchase_amount, Spender Category
86 SELECT CUSTOMER_ID,
87 COALESCE(PURCHASE_AMOUNT,0) AS PURCHASE_AMOUNT,
88 CASE WHEN PURCHASE_AMOUNT > 80 THEN 'HIGH'
89 WHEN PURCHASE_AMOUNT BETWEEN 50 AND 80 THEN 'MEDIUM'
90 ELSE 'LOW'
91 END AS SPEND_CATEGORY
92 FROM SHOPPING_TRENDS_DATASET;
93

```

Results (just now)

Table Chart 300 rows 59ms

#	CUSTOMER_ID	PURCHASE_AMOUNT	SPEND_CATEGORY
1	1	20	LOW
2	2	21	LOW
3	3	27	LOW
4	4	45	LOW
5	5	80	MEDIUM
6	6	82	HIGH
7	7	50	MEDIUM
8	8	29	LOW

Feedback

8. Find customers who have no Previous Purchases value but whose Color is not NULL. Expected Columns: Customer ID, Color, Previous Purchases

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```

93
94 --8. Find customers who have no Previous
95 --Purchases value but whose Color is not NULL.
96 --Expected Columns: Customer ID, Color, Previous Purchases
97 SELECT CUSTOMER_ID,
98 COLOR,
99 PREVIOUS_PURCHASES
100 FROM SHOPPING_TRENDS_DATASET
101 WHERE COLOR IS NOT NULL;
102
103 --9. Group records by Frequency of
104 --Purchases and show the total amount spent per group, treating NULL frequenc
105 --ies as 'Unknown'.

```

Results (just now)

Table Chart 271 rows 65ms

#	CUSTOMER_ID	COLOR	PREVIOUS_PURCHASES
1	2	Turquoise	12
2	3	Black	50
3	4	Green	11
4	5	Turquoise	19
5	6	White	36
6	7	Yellow	25
7	8	Green	null
8	9	Pink	39

Feedback

9. Group records by Frequency of Purchases and show the total amount spent per group, treating NULL frequencies as 'Unknown'. Expected Columns: Frequency of Purchases, Total purchase_amount

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```

102
103 --9. Group records by Frequency of
104 --Purchases and show the total amount spent per group, treating NULL frequenc
105 --ies as 'Unknown'.
106 --Expected Columns: Frequency of Purchases, Total purchase_amount
107 SELECT COALESCE(FREQUENCY_OF_PURCHASES, 'UNKNOWN') FREQUENCY_OF_PURCHASES,
108 SUM(PURCHASE_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
109 FROM SHOPPING_TRENDS_DATASET
110 GROUP BY FREQUENCY_OF_PURCHASES;
111
112
113
114
115
116

```

Results (1 minute ago)

Table Chart 271 rows 65ms

#	CUSTOMER_ID	COLOR	PREVIOUS_PURCHASES
1	2	Turquoise	12
2	3	Black	50
3	4	Green	11
4	5	Turquoise	19
5	6	White	36
6	7	Yellow	25
7	8	Green	null
8	9	Pink	39

Feedback

10. Display a list of all Category values with the number of times each was purchased, excluding rows where Category is NULL. Expected Columns: Category, Total Purchases

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```

117
118 --10. Display a list of all Category values with the number of times each was purcha
119 --sed, excluding rows where Category is NULL.
120 --Expected Columns: Category, Total Purchases
121 SELECT CATEGORY,
122 COUNT(*) AS TIMES_PURCHASED,
123 FROM SHOPPING_TRENDS_DATASET
124 WHERE CATEGORY IS NOT NULL
125 GROUP BY CATEGORY;
126
127
128
129
130
131 --11. Return the top

```

Results (just now)

Table Chart 4 rows 88ms

#	CATEGORY	TIMES_PURCHASED
1	Outerwear	60
2	Footwear	70
3	Clothing	59
4	Accessories	78

Feedback

11. Return the top 5 Locations with the highest total purchase_amount, replacing NULLs in amount with 0. Expected Columns: Location, Total purchase_amount

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```

129
130
131 --11. Return the top
132 --5 Locations with the highest total purchase_amount, replacing NULLs in amount
133 --with 0.
134 --Expected Columns: Location, Total purchase_amount
135 SELECT LOCATION,
136 SUM (COALESCE(PURCHASE_AMOUNT, '0')) AS TOTAL_PURCHASE_AMOUNT,
137 FROM SHOPPING_TRENDS_DATASET
138 GROUP BY LOCATION
139 ORDER BY TOTAL_PURCHASE_AMOUNT DESC
140 LIMIT 1;
141
142
143

```

Results (3 minutes ago)

Table Chart 1 row 80ms

	LOCATION	# TOTAL_PURCHASE_AMOUNT
1	Maine	2294

Feedback

12. Group customers by Gender and Size, and count how many entries have a NULL Color. Expected Columns: Gender, Size, Null Color Count

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```

144
145
146 --12. Group customers by Gender and Size, and count how many entries have a NULL
147 -- L Color.
148 --Expected Columns: Gender, Size, Null Color Count
149 SELECT GENDER,
150 SIZE,
151 COUNT(COLOR) AS NULL_COLOR_COUNT
152 FROM SHOPPING_TRENDS_DATASET
153 WHERE COLOR IS NULL
154 GROUP BY GENDER, SIZE;
155
156 --13. Identify all Item Purchased where more than 3 purchases had NULL Shipping
157 --Type.
158 --Expected Columns: Item Purchased, NULL Shipping Type Count

```

Results (just now)

Table Chart 5 rows 89ms

	GENDER	SIZE	# NULL_COLOR_COUNT
1	Male	null	0
2	Male	M	0
3	Male	L	0
4	Male	S	0
5	Male	XL	0

Feedback

13. Identify all Item Purchased where more than 3 purchases had NULL Shipping Type. Expected Columns: Item Purchased, NULL Shipping Type Count

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```

153 WHERE COLOR IS NULL
154 GROUP BY GENDER, SIZE;
155 -----
156 --13. Identify all Item Purchased where more than 3 purchases had NULL Shipping
157 --Type.
158 --Expected Columns: Item Purchased, NULL Shipping Type Count
159 SELECT ITEM_PURCHASED,
160 COUNT(*) AS NULL_SHIPPING_TYPE
161 FROM SHOPPING_TRENDS_DATASET
162 GROUP BY ITEM_PURCHASED
163 HAVING COUNT(*) >3;
164 -----
165
166
167
168

```

Results (just now)

Table Chart 11 rows 78ms

	ITEM_PURCHASED	# NULL_SHIPPING_TYPE
1	Shirt	26
2	Jeans	25
3	Sweater	27
4	Coat	21
5	Sandals	32
6	Sneakers	39
7	Handbag	20
8	Shoes	26

Feedback

14. Show a count of how many customers per Payment Method have NULL Review Rating. Expected Columns: Payment Method, Missing Review Rating Count

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```

168
169
170 --14. Show a count of how many customers per Payment Method have NULL Review
171 --Rating.
172 --Expected Columns: Payment Method, Missing Review Rating Count
173 SELECT PAYMENT_METHOD,
174 COUNT(REVIEW_RATING) AS MISSING_REVIEW_RATING_COUNT
175 FROM SHOPPING_TRENDS_DATASET
176 WHERE REVIEW_RATING IS NULL
177 GROUP BY PAYMENT_METHOD;
178 -----
179
180
181
182
183

```

Results (just now)

Table Chart 7 rows 76ms

	PAYMENT_METHOD	# MISSING_REVIEW_RATING_COUNT
1	Credit Card	0
2	Cash	0
3	null	0
4	Debit Card	0
5	Venmo	0
6	PayPal	0
7	Bank Transfer	0

Feedback

15. Group by Category and return the average Review Rating, replacing NULLs with 0, and filter only where average is greater than 3.5. Expected Columns: Category, Average Review Rating

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```

180
181
182
183
184
185 --15. Group by Category and return the average Review Rating, replacing NULLs with
186 --0, and filter only where average is greater than 3.5.
187 --Expected Columns: Category, Average Review Rating
188 SELECT CATEGORY,
189 AVG(COALESCE(REVIEW_RATING,0)) AS AVERAGE_REVIEW_RATING
190 FROM SHOPPING_TRENDS_DATASET
191 GROUP BY CATEGORY
192 HAVING AVG(COALESCE(REVIEW_RATING,0)) >3.5;
193 -----
194 --16. List all Colors that are missing (NULL) in at least
195 --2 rows and the average Age of customers for those rows.

```

Results (2 minutes ago)

Table Chart 0 rows 87ms

CATEGORY	AVERAGE_REVIEW_RATING
Query produced no results	

Feedback

16. List all Colors that are missing (NULL) in at least 2 rows and the average Age of customers for those rows. Expected Columns: Color, Average Age

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```

191 GROUP BY CATEGORY
192 HAVING AVG(COALESCE(REVIEW_RATING,0)) >3.5;
193 -----
194 --16. List all Colors that are missing (NULL) in at least
195 --2 rows and the average Age of customers for those rows.
196 --Expected Columns: Color, Average Age
197 SELECT COLOR,
198 AVG(AGE) AS AVERAGE_AGE
199 FROM SHOPPING_TRENDS_DATASET
200 WHERE COLOR IS NULL
201 GROUP BY COLOR;
202 -----
203
204
205
206

```

Results (just now)

Table Chart 1 row 87ms

COLOR	AVERAGE_AGE
1 null	47.846154

Feedback

17. Use CASE to create a column Delivery Speed: 'Fast' if Shipping Type is 'Express' or 'Next Day Air', 'Slow' if 'Standard', 'Other' for all else including NULL. Then count how many customers fall into each category.

Home ... x 2025-10-16 9:53pm 2025-10-16 9:55pm 2025-10-18 10:19pm +

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```
--18. Find customers whose purchase_amount is NULL and whose Promo Code Used is
--'Yes'.
--Expected Columns: Customer ID, purchase_amount, Promo Code Used -
SELECT CUSTOMER_ID,
PURCHASE_AMOUNT,
PROMO_CODE_USED
FROM SHOPPING_TRENDS_DATASET
WHERE PURCHASE_AMOUNT IS NULL AND PROMO_CODE_USED = 'YES';
-----
--19. Group by Location and show the maximum Previous
```

Results (just now)

Table Chart 20 rows 73ms

	# CUSTOMER_ID	# PURCHASE_AMOUNT	0 1 PROMO_CODE_USED
1		13	null
2		30	null
3		78	null
4		95	null
5		124	null
6		129	null
7		130	null
8		138	null

Feedback

18. Find customers whose purchase_amount is NULL and whose Promo Code Used is 'Yes'. Expected Columns: Customer ID, purchase_amount, Promo Code Used

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```
--19. Group by Location and show the maximum Previous
--Purchases, replacing NULLs with 0, only where the average rating is above 4.0.
--Expected Columns: Location, Max Previous Purchases, Average
--Review Rating
SELECT LOCATION,
MAX(COALESCE(PREVIOUS_PURCHASES,0))
AS MAXIMUM_PREVIOUS_PURCHASES,
AVG(REVIEW_RATING) AS AVERAGE_RATING
FROM SHOPPING_TRENDS_DATASET
GROUP BY LOCATION
HAVING AVERAGE_RATING > 4.0;
-----
```

Results (just now)

Table Chart 0 rows 88ms

LOCATION	MAXIMUM_PREVIOUS_PURCHASES	AVERAGE_RATING
Query produced no results		

Feedback

19. Group by Location and show the maximum Previous Purchases, replacing NULLs with 0, only where the average rating is above 4.0. Expected Columns: Location, Max Previous Purchases, Average Review Rating

```

226
227
228 --19. Group by Location and show the maximum Previous
229 --Purchases, replacing NULLs with 0, only where the average rating is above 4.0.
230 --Expected Columns: Location, Max Previous Purchases, Average
231 --Review Rating
232 SELECT LOCATION,
233 MAX(COALESCE(PREVIOUS_PURCHASES,0))
234 AS MAXIMUM_PREVIOUS_PURCHASES,
235 AVG(REVIEW_RATING) AS AVERAGE_RATING
236 FROM SHOPPING_TRENDS_DATASET
237 GROUP BY LOCATION
238 HAVING AVERAGE_RATING > 4.0;
239
240
241

```

Results (26 minutes ago)

Table Chart 0 rows 88ms

LOCATION	MAXIMUM_PREVIOUS_PURCHASES	AVERAGE_RATING
Query produced no results		

Feedback

20. Show customers who have a NULL Shipping Type but made a purchase in the range of 30 to 70 USD. Expected Columns: Customer ID, Shipping Type, purchase_amount, Item Purchased

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```

241
242
243 --20. Show customers who have a NULL Shipping
244 --Type but made a purchase in the range of 30 to 70 USD.
245 --Expected Columns: Customer ID, Shipping
246 --Type, purchase_amount, Item Purchased
247 SELECT CUSTOMER_ID,
248 SHIPPING_TYPE,
249 PURCHASE_AMOUNT,
250 ITEM_PURCHASED
251 FROM SHOPPING_TRENDS_DATASET
252 WHERE PURCHASE_AMOUNT BETWEEN 30 AND 70
253 AND SHIPPING_TYPE IS NULL;
254
255

```

Results (29 minutes ago)

Table Chart 0 rows 88ms

LOCATION	MAXIMUM_PREVIOUS_PURCHASES	AVERAGE_RATING
Query produced no results		

Feedback